

Using statistical learning models with household survey microdata

Applications to energy demand, energy poverty, and environmental taxation

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EAERE

European Association
of Environmental and
Resource Economists

Recent applications of AI in Environmental and
Resource Economics



Session 2

Modelling approaches and Statistical learning models

Prediction is not causal inference

A necessary distinction for applied environmental economists

Prediction

- Predict outcomes for households not used to train the model.
- Useful for targeting, imputation and risk ranking.
- Evaluated with out-of-sample metrics such as AUC or RMSE.

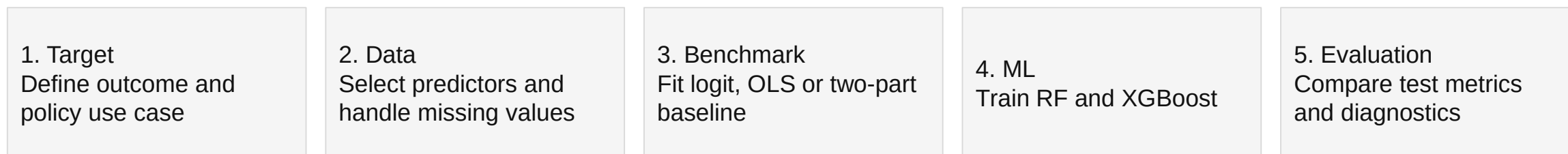
Causal inference

- Estimate the effect of a policy, price change or intervention.
- Requires an identification strategy beyond the predictive model.
- High predictive performance does not establish causality.

Course focus *Prediction and policy design with household survey microdata; causal claims require a separate design.*

Applied statistical-learning workflow

The same logic applies to energy poverty, energy demand and transport fuels



- Preprocessing is learned inside recipes/pipelines using training data only.
- Variables that mechanically define the target are excluded to avoid leakage.
- Predictions are translated back into weighted population-level summaries.

Why keep a benchmark model?

The benchmark protects us from over-selling complexity

- Binary outcomes: logit or probit provides a transparent baseline.
- Continuous outcomes: OLS or log-linear models establish a simple reference point.
- Value added I: coefficients force users to check signs, scales and plausibility.
- Value added II: readers see whether ML improves prediction or only adds complexity.

Decision rule *If a flexible model does not clearly improve predictive performance, the simpler model may be preferred for communication.*

Random Forest and XGBoost

Two tree-based models for nonlinear relationships in tabular microdata

Random Forest

- Averages many trees built on random subsets of rows and predictors.
- Robust default model with moderate tuning needs.
- Captures nonlinearities and interactions automatically.

XGBoost

- Builds trees sequentially; each new tree corrects previous errors.
- Often strong on structured tabular data.
- Requires more careful tuning and interpretation.

Feature	Random Forest	XGBoost
Nonlinear effects	Yes	Yes
Interactions	Automatic	Automatic
Tuning need	Moderate	Higher
Interpretation	Importance + profiles	Importance + profiles

Preprocessing survey microdata for ML

Most errors happen before the algorithm is called

- Missing values: impute inside the recipe or pipeline, not before splitting the data.
- Categorical predictors: encode factors consistently across cross-validation folds.
- Leakage risk: do not use energy_share to predict energy_poor_10pct; it defines the target.
- Policy realism: predictors should match variables actually available for targeting or forecasting.

Cross-validation and tuning

Useful, but not the whole story

1. Train/test split

Held-out test data estimate final performance

2. Cross-validation

Training data are split into folds to tune parameters

3. Final test

Used once at the end; never tune on it

RF tuning

- mtry
- minimum node size
- number of trees

XGBoost tuning

- tree depth and learning rate
- loss reduction and subsampling
- mtry / column sampling

Metrics for classification

Energy poverty yes/no

Metric	What it answers	Policy interpretation
ROC AUC	Does the model rank vulnerable households higher?	Threshold-free targeting potential
Sensitivity	What share of vulnerable households are detected?	Coverage / exclusion errors
Specificity	What share of non-vulnerable households are excluded?	Administrative efficiency

Metrics for regression

Home energy expenditure, electricity demand and fuel spending

Metric	Meaning	How to read it
RMSE	Root mean squared error	Penalises large mistakes strongly
MAE	Mean absolute error	Average prediction error in euros or kWh
R ²	Share of variance explained	Useful summary, but not enough alone

Hands-on 2: Modelling energy demand and energy poverty

Hands-on 2_01 — machine learning models

Random forest and XGBoost

- **Classification - energy_poor_10pct**
- **Regression - home_energy_spend_eur**

Gas demand motivates a two-part model

- Zeros are not random noise. They reflect lack of gas access, use of alternative heating systems, warm climates or household technology choices.
- A single continuous regression on all households tends to blur two different decisions: participation and intensity.
- The two-part model separates the extensive margin (does the household consume gas?) from the intensive margin (how much does it consume if it consumes?).
- Part I is a binary classification problem: estimate the probability that household h consumes fuel e in period t .
- Part II is a continuous regression problem: estimate consumption conditional on positive consumption.
- The unconditional expected demand is the product of the two predictions: predicted probability of use times predicted conditional consumption.

$$\begin{aligned}d_i &= 1[y_i > 0] \\ E(y_i | X_i) &= \Pr(d_i = 1 | X_i) \times E(y_i | d_i = 1, X_i) \\ \hat{y}_i &= \hat{p}_i \times \hat{g}_i\end{aligned}$$

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- **Model the gas expenditure using a two-part model approach**